2110201: COMP ENG MATH (2023/2) Quiz 3

Name: Student ID: Entry Number:

1. (4 points) Find the determinant of the matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ -2 & -1 & 2 & -3 \\ 3 & 0 & 1 & x \end{bmatrix}$. where x represents the 8th digit in your student ID. For example, if your ID is 6531234**5**21, then x = 5 for your matrix.
[**Hint:** The answer is an integer.]

2. (4 points) Find the inverse (if any) of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 6 & 9 \end{bmatrix}$.

3. (2 points, 0.5 points each) Given a matrix A with the properties in each sub-question.
Is the matrix A invertible? (Answer **Yes** or **No** in the table provided.)

3.1	**3.2**	**3.3**	**3.4**

3.1. $\mathbf{A} = \mathbf{I}_4$

3.2. $\text{RRE(A)} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$

3.3. $\mathbf{A}_{3\times 3}$ and there exists a matrix $\mathbf{B}$ such that $\mathbf{BA} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

3.4. $\mathbf{A}^{2024} = \mathbf{I}_{2567}$